

Two-Stage Geometric Audit for Malicious Client Detection

A geometric view of the audit logic — Stage 1 rejects directionally weak or negatively aligned clients using ϵ_{dir} , while Stage 2 applies robust median/MAD magnitude screening to the provisional pass set and returns accepted encrypted submissions $\mathcal{A}^{(t)}$ and rejected tags $\mathcal{D}^{(t)}$.

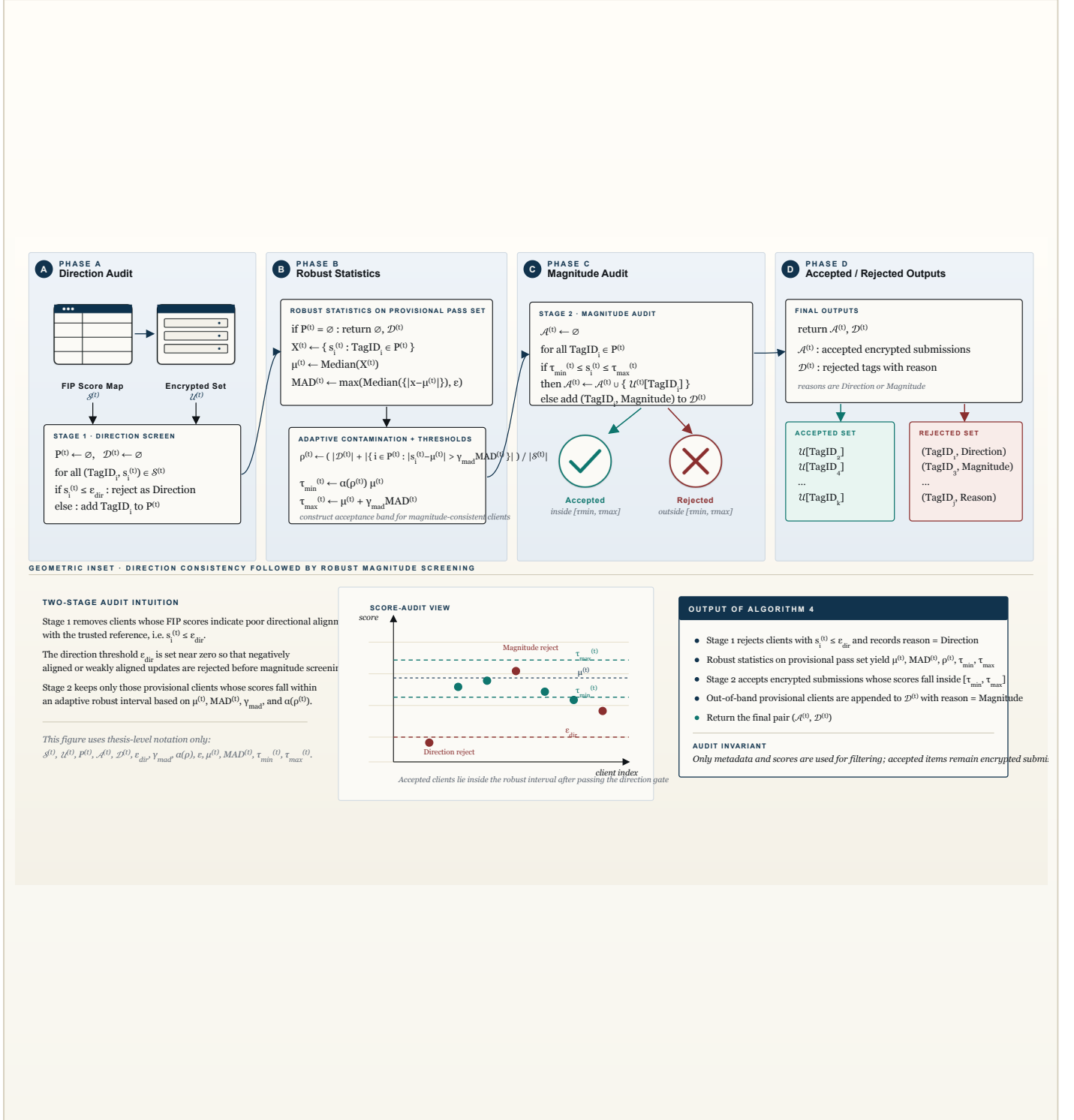


Figure 4. Algorithm 4. The verifier takes the FIP score map $\mathcal{S}^{(t)}$ and encrypted submission set $\mathcal{U}^{(t)}$. In Stage 1, it rejects clients with $s_i^{(t)} \leq \epsilon_{\text{dir}}$ as directionally suspicious and forms the provisional pass set $P^{(t)}$. In Stage 2, it computes the robust statistics $\mu^{(t)}$ and $\text{MAD}^{(t)}$, estimates the contamination rate $\rho^{(t)}$, and constructs the adaptive interval $[\tau_{\min}^{(t)}, \tau_{\max}^{(t)}]$. Encrypted submissions whose scores fall inside this band are retained in the accepted set $\mathcal{A}^{(t)}$, while the others are added to the rejected tag set $\mathcal{D}^{(t)}$ with reason Direction or Magnitude. The inset visualizes this two-stage screening in score space.